Question

Tell me about a time you failed, or a project that didn't go as planned.

Answer

A meaningful failure for me was a personal project called Cirrus, where I set out to build an AI-driven system to predict wildfires on a map across Canada. The idea was to use free historical weather data from NOAA, but I ran into a fundamental data problem I couldn't solve.

The data came from weather stations scattered inconsistently across Canada—some areas like the tundra had zero coverage, and in remote regions stations were sometimes tens of kilometres apart or more. I tried to interpolate the data onto a grid to fill the gaps, but that meant some tiles were essentially guesses that weren't reliable enough for something that might influence real wildfire-risk decisions. At that point I had to accept the project wasn't going to work and stop rather than putting out something people couldn't trust.

I took two big lessons from that. First, I need to validate my assumptions about data quality and coverage early, not after I've built half the system. Second, I need clear criteria for when to stop so I'm not just following sunk costs forward. Those lessons directly influenced how I approached later projects like the self-teaching recommendation system and my portfolio chatbot: I now prototype on small, representative data, measure whether it's actually good enough for the use case, and only scale once I know the foundations are solid. That's the kind of early validation I'd encourage on any team project—making sure the data constraints are clear before we commit resources.

Concepts to memorise

1. The failure
   - Personal project: Cirrus, an AI-driven wildfire prediction system for Canada that failed.
   - Idea: use free historical NOAA weather data to visualise wildfire risk.
   - Ran into a fundamental data problem I couldn't solve.
2. The fundamental data problem
   - Weather stations scattered inconsistently across Canada; some areas (e.g., tundra) had zero coverage.
   - In remote regions, stations were sometimes tens of kilometres apart or more.
   - Tried to interpolate data onto a grid to fill gaps.
   - Result: some tiles were unreliable guesses—not good enough for wildfire-risk decisions.

3. The decision
    - Had to accept the data problem was fundamental and the project wasn't going to work.
    - Chose to stop rather than putting out something people couldn't trust.
4. Lessons and changes
    - Validate data quality and coverage early, not after building half the system.
    - Define clear stop criteria to avoid sunk-cost thinking.
    - Apply those lessons to later AI/RAG projects: prototype on small representative data, measure quality against the use case, only scale once foundations are solid.
    - That's the kind of early validation you'd encourage on any team project—making sure constraints are clear before committing resources.